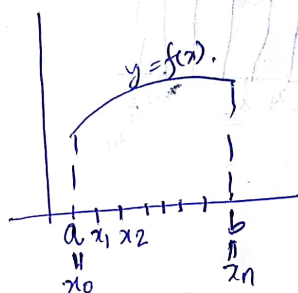


Numerical Integration

$x_0, x_1, x_2, \dots, x_n$

→ Here we take the interval as $a = x_0, x_1 = x_0 + h$
 $x_2 = x_0 + 2h, \dots, x_n = x_0 + nh$.



$$\therefore I = \int_{x_0}^{x_n} f(x) dx = \int_{x_0}^{x_0+nh} f(x) dx$$

put $x = x_0 + uh$

limits, when $x = x_0$
 $u = 0$

where $x = x_0 + uh$
 then $u = n$.

Subst. all these in (1).

$$I = \int_0^n f(x_0 + uh) h du$$

now we are expanding $f(x_0 + uh)$ by using Newton's forward difference interpolation polynomial. & then integrate term by term. we are getting an expression.

$$= h \cdot \int_0^n y_0 du + \int_0^n \frac{u}{1!} \Delta y_0 + \int_0^n \frac{u(u-1)}{2!} \Delta^2 y_0 + \dots$$

$$= h \left(y_0 + \frac{\Delta^2}{2} y_0 + \frac{\frac{n^3}{3} - \frac{n^2}{2}}{2!} \Delta^2 y_0 + \dots \right) \rightarrow (2)$$

Trapezoidal Rule :-

put $n=1$ in eq (2)

we have $\rightarrow$
 $\int_{x_0}^{x_0+h} f(x) dx =$

$$h \left\{ y_0 + \frac{1}{2} y_1 \right\}$$

$$= h \left\{ y_0 + \frac{1}{2} (y_1 - y_0) \right\}$$

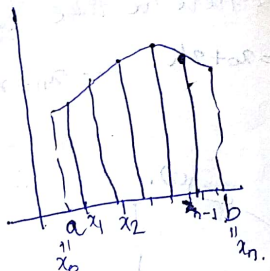
$$= h \cdot \frac{2y_0 + y_1 - y_0}{2}$$

$$= \frac{h}{2} \{ y_0 + y_1 \}$$

consider the second trapezium in the interval x_1 to x_2

$$\therefore \int_{x_1}^{x_0+2h} f(x) dx = \frac{h}{2} [y_1 + y_2]$$

$$\therefore \int_{x_0}^{x_n} f(x) dx = \frac{h}{2} [y_0 + y_1 + y_1 + y_2 + \dots + y_{n-1} + y_n]$$



it can be seen that $\rightarrow$
 all the ~~terms~~ except I^1 last one repeating hence
 the formula can be written as $\rightarrow$

$$I = \frac{h}{2} [y_0 + y_n + 2(y_1 + y_2 + \dots + y_{n-1})]$$

Simpson's $\frac{1}{3}$ rd rule :-

put $n=2$ in eq. no. (2).

$$\int_{x_0}^{x_0+2h} f(x) dx = \frac{h}{3} \left[(y_0 + y_2) + 2(y_1) \right]$$

Simpson's $\frac{3}{8}$ th rule :-

put $n=3$ in eq (2)

$$\int_{x_0}^{x_0+3h} f(x) dx = \frac{3h}{8} \left[(y_0 + y_3) + 3(y_1 + y_2) + 2(y_1 + y_2 + y_3) \right]$$

Q. evaluate.

$$\int_0^1 \frac{dx}{1+x^2}$$

taking $h = \frac{1}{4}$ using Simpson's $\frac{1}{3}$ rule
hence compute an
approximate value of π .

$$h = \frac{1}{4}, \quad n = \frac{b-a}{h} = \frac{1-0}{\frac{1}{4}} = 4.$$

~~$y = \frac{1}{1+x^2}$~~

$$I = \frac{h}{3} \left[(y_0 + y_n) + 2(y_1 + y_2 + \dots) + 4(y_1 + y_3 + y_5) \right]$$

x	$x_0 = 0$	$x_1 = \frac{1}{4}$	$x_2 = \frac{2}{4}$	$x_3 = \frac{3}{4}$	$x_4 = \frac{4}{4} = 1$
$y = \frac{1}{1+x^2}$	$\frac{1}{1+0^2} = 1$	$\frac{1}{1+(\frac{1}{4})^2}$	$\frac{1}{1+(\frac{2}{4})^2}$	$\frac{1}{1+(\frac{3}{4})^2}$	$\frac{1}{1+1} = \frac{1}{2}$

$$\int_0^1 \frac{dx}{1+x^2} = 0.785$$

$$(\tan^{-1}x)' = 0.785$$

$$\Rightarrow \frac{\pi}{4} - 0 = 0.785$$

$$\Rightarrow \pi = 0.785 \times 4$$

$$= 3.14$$

Q: Evaluate $\int_0^4 e^x dx$ by using Simpson's $\frac{1}{3}$ rd rule

given that

$$e = 2.72$$

$$e^2 = 7.39$$

$$e^3 = 20.09$$

$$e^4 = 54.6$$

$$n = \frac{b-a}{h}$$

$$h = \frac{4-0}{4} = 1$$

x	0	1	2	3	4
y	e ⁰	e ¹	e ²	e ³	e ⁴

$$\frac{1}{3}rd = \frac{h}{3} \{ (y_0 + y_n) + 2(y_2 + y_4) + 4(y_1 + y_3) \}$$

$$= \frac{1}{3} \{ (1 + e^4) + 2(e^2 + 54.6) + 4(e + 20.09) \}$$

$$= \frac{1}{3} \{ 55.6 + 40.18 + 91.24 \}$$

$$\frac{91.24 + 40.18}{55.6} = 2.72$$

$$= \frac{187.02}{3} = 62.34$$

Q: Vel. of moped starts from rest is given at fixed interval of time 't'. as given below

t	0	2	4	6	8	10	12	14	16	18	20
v	0	10	18	25	29	32	20	11	5	2	0

Estimate approximately the dist. covered in 20 min.

we have $\frac{ds}{dt} = v$ or $ds = v dt$

$$\int_0^{20} ds = \int_0^{20} v dt$$

we are dividing the values as $v_0, v_1, v_2, \dots, v_{10}$ then integration by using Simpson's rule (any formula can be applied)

$$\therefore \int_0^{20} ds = \frac{h}{3} \{ (v_0 + v_{10}) + 2(v_2 + v_4 + v_6 + v_8 + v_{10}) + 4(v_1 + v_3 + v_5 + v_7 + v_9) \}$$

$$= 309.3?$$

Q. No

Evaluate the integral
Simpson's $\frac{1}{3}$ rd & $\frac{3}{8}$ th rule

$$\int_4^{5.2} \log_e x \, dx \text{ using}$$

Hint: Here divide the interval into 6 intervals

Q: The vel. of a particle at a distance s from a point on its path is given by the table $\rightarrow$

s	0	10	20	30	40	50	60	70
v	47	58	64	65	61	52	38	20

Estimate the time taken to cover 60 feet by using Simpson's rule.

Q. No

$$t = 20 \quad \frac{ds}{dt} = v$$

$$v dt = ds$$

$$dt = \frac{ds}{v}$$

$$(t)_0^{60} = \frac{1}{3} \left(\left(\frac{1}{v_0} + \frac{1}{v_6} \right) + 2 \left(\frac{1}{v_2} + \frac{1}{v_4} \right) + 4 \left(\frac{1}{v_1} + \frac{1}{v_3} + \frac{1}{v_5} \right) \right)$$

Ans.

$$y_1 = y_0 + \frac{h}{1!} y_0' + \frac{h^2}{2!} y_0'' + \dots \quad (\because x_1 - x_0 = h)$$

Similarly $y_2 = y_1 + \frac{h}{1!} y_1' + \frac{h^2}{2!} y_1'' + \dots$

In general,

$$y_{n+1} = y_n + \frac{h}{1!} y_n' + \frac{h^2}{2!} y_n'' + \dots$$

prob. solve, $\frac{dy}{dx} = x+y$ given $y(1)=0$ & get ~~$y(1.1)$~~

$y(1.1)$, $y(1.2)$ by Taylor series method.

sol.

$$y_{n+1} = y_n + \frac{h}{1!} y_n' + \frac{h^2}{2!} y_n'' + \dots$$

~~$y(1.1)$~~

~~$x_0 = 1$~~ , $x_0 = 1$
 $y_0 = 0$

$$y_1 = y(x_1) = y(1.1) = ?$$

$$y_2 = y(x_2) = y(1.2) = ?$$

$x_0 = 1$	$x_1 = 1.1$	$x_2 = 1.2$
$y_0 = 0$	$y_1 = ?$	$y_2 = ?$

$$h = 1.1 - 1 = 0.1$$

$$y_{n+1} = y_n + \frac{h}{1!} y'_n + \frac{h^2}{2!} y''_n + \dots$$

Given $n=0$ & $n=1$
to get y_1 and y_2 .

$$y_1 = y_0 + \frac{h}{1!} y'_0 + \frac{h^2}{2!} y''_0 + \dots$$

$$= 0 + \frac{0.1}{1} \times 1 + \frac{(0.1)^2}{2} \times 2 + \frac{(0.1)^3}{3!} \times 2 + \frac{(0.1)^4}{4!} \times 2$$

$$= 0.1 + 0.01 + \dots$$

$$= 0.11033847$$

$n=1$,

$$y_2 = y_1 + \frac{h}{1!} y'_1 + \frac{h^2}{2!} y''_1 + \frac{h^3}{3!} y'''_1$$

$$= 0.11033847 + \frac{0.1}{1} \times 1.21033847$$

$$+ \frac{(0.1)^2}{2} \times 2.21033847$$

$$+ \frac{(0.1)^3}{3!} \times 2.21033847$$

$$y_1''' = y_1''$$

$$= 2.21033847$$

$$y'_1 = x + y_1 = 1.1 + 0.11033847$$

$$y''_1 = 1 + y'_1$$

$$= 2.21033847$$

~~0.11033847~~

~~0.11033847~~

0.2461072

using Taylor series method find $y(0.2)$ & $y(0.4)$ correct to 4 decimal places given $\frac{dy}{dx} = 1 - 2xy$ and $y(0) = 1$

$x_0 = 0$	$x_1 = 0.2$	$x_2 = 0.4$
$y_0 = 1$	$y_1 = ?$	$y_2 = ?$

$$y_{n+1} = y_n + \frac{h}{1!} y'_n + \frac{h^2}{2!} y''_n + \frac{h^3}{3!} y'''_n + \dots$$

$$y_1 = y_0 + \frac{h}{1!} y'_0 + \frac{h^2}{2!} y''_0 + \frac{h^3}{3!} y'''_0 + \dots$$

$$y_1 = 1 + 0.2 \times 1 + \frac{(0.2)^2}{2} \times -2 + \frac{(0.2)^3}{3!} \times -4$$

$$= 1.165333333$$

$$y'_1 = 1 - 2xy$$

$$y'_0 = 1 - 2x_0 y_0$$

$$= 1 - 2 \times 0$$

$$= 1$$

$$y''_0 = -2[x_0 y'_0 + y_0]$$

$$= -2[1]$$

$$y'''_0 = -2(x_0 y''_0 + y'_0)$$

$$= -2[1 - 2 \times 1]$$

$$= -2[-1]$$

$$= 2$$

$$(y(0.2)) + y(0.4) = 1.165333333 + 0.2461072 = 1.411440533$$